

Abstract

Child malnutrition remains a serious public health concern in India, with large differences observed across districts. Among the key indicators of malnutrition, child stunting and child wasting reflect long-term and short-term nutritional deprivation, respectively. This study aims to predict district-level prevalence of child stunting and wasting using machine learning models based on NFHS-5 (2019–20) District Fact Sheet data covering 706 districts and 109 indicators. After data preprocessing and feature selection using correlation analysis and LASSO regression, two separate predictive models were developed for stunting and wasting. Linear Regression, Random Forest and XGBoost models were applied and evaluated using MAE, RMSE and R^2 scores. Further, Particle Swarm Optimization (PSO) was used to tune the hyperparameters of the Random Forest model. The results show that wasting was predicted with higher accuracy than stunting, with the best test R^2 of about 0.78, while stunting achieved a best test R^2 of about 0.63. PSO slightly improved the performance of the Random Forest model. Maternal nutrition, sanitation, women's education and child anaemia emerged as important predictors. The findings highlight the usefulness of machine learning methods for district-level monitoring of child malnutrition and for supporting targeted policy interventions

Keywords

Child malnutrition, Stunting, Wasting, Machine learning, NFHS-5, Particle Swarm Optimization

Introduction

Child malnutrition continues to be one of the most serious public health challenges in India, despite several decades of policy efforts and nutrition programmes. Malnutrition in early childhood affects physical growth, brain development, learning ability and long-term health

outcomes. Among the different forms of malnutrition, child stunting (low height for age) reflects long-term nutritional deprivation, while child wasting (low weight for height) indicates acute undernutrition. Both conditions increase the risk of illness, mortality and poor cognitive development among children.

According to recent national surveys, the burden of stunting and wasting remains high in many parts of the country, with large variations across states and districts. These variations reflect differences in maternal health, sanitation, food security, healthcare access and socio-economic conditions. Therefore, understanding the district-level patterns of child malnutrition is essential for effective planning and targeted policy intervention.

India shows significant regional diversity in terms of living conditions, health infrastructure and social development. National and state-level averages often hide large inequalities that exist within states. District-level analysis helps in identifying high-burden areas more precisely and allows policymakers to design location-specific interventions. However, most existing studies on child malnutrition rely on individual or household-level survey data, while fewer studies focus on predictive modelling at the district level.

With the availability of detailed district-level data from the National Family Health Survey (NFHS-5), it is now possible to apply data-driven methods to study the spatial distribution of child malnutrition and its key determinants.

Traditional statistical methods have been widely used to study the determinants of child malnutrition. While these methods are useful for interpretation and causal analysis, they may not fully capture complex non-linear relationships among a large number of predictors. Machine learning (ML) techniques offer an alternative approach by improving predictive accuracy when dealing with high-dimensional data, interactions among variables and non-linear effects.

In recent years, ML has been increasingly used in public health research for disease prediction, risk classification and health outcome forecasting. However, the application of ML for district-level prediction of child stunting and wasting in India remains limited.

Most previous studies on child malnutrition in India:

- Focus on individual- or household-level microdata,
- Use conventional regression-based methods and
- Rarely explore district-level predictive modelling using machine learning.

Moreover, very few studies apply meta-heuristic optimization techniques, such as Particle Swarm Optimization (PSO), to improve the performance of machine learning models in nutritional outcome prediction.

The present study aims to fill these gaps by developing a district-level predictive framework for child malnutrition using machine learning and optimization techniques. The specific objectives of the study are:

To predict district-level child stunting (%) using machine learning models based on NFHS-5 data.

To predict district-level child wasting (%) using machine learning models.

To identify the most important predictors associated with stunting and wasting through feature selection and model interpretation.

To improve model performance using Particle Swarm Optimization (PSO) for hyperparameter tuning.

By combining public health insights with machine learning methods, this study seeks to provide evidence that can support better district-level monitoring and targeted nutrition policies in India.

Literature Review

Child malnutrition has been widely studied in India due to its long-standing impact on child health, survival and human capital formation. Among the different forms of malnutrition, stunting and wasting have received particular attention because they reflect chronic and acute nutritional deprivation, respectively. Earlier studies based on the National Family Health Survey (NFHS) rounds have consistently shown that India continues to face a high burden of both stunting and wasting, with strong regional inequalities across states and districts.

Several studies have identified maternal education, maternal nutrition, household sanitation, drinking water access and economic conditions as major determinants of child stunting. Higher levels of women's education have been found to improve child nutrition outcomes through better health awareness, improved feeding practices and greater use of healthcare services. Poor sanitation and open defecation have also been shown to contribute significantly to chronic undernutrition by increasing repeated infections and poor nutrient absorption among children.

In contrast, studies on child wasting highlight the role of short-term health and nutritional shocks, such as childhood illnesses, diarrhoea, fever and inadequate complementary feeding practices. Acute food insecurity and seasonal variations have also been linked with higher wasting prevalence. Unlike stunting, wasting responds more quickly to recent changes in diet and disease environment. Recent research has also emphasized the strong association between child anaemia, maternal anaemia and wasting, indicating the link between micronutrient deficiencies and acute malnutrition.

With the growing availability of large health datasets, researchers have started using machine learning methods to improve the prediction of nutritional outcomes. Some studies have applied decision trees, random forests and boosting techniques to predict child malnutrition using household survey data and have reported improved predictive performance compared to traditional regression models. These studies show that machine learning can capture complex non-linear relationships among socio-economic, maternal and environmental factors.

However, most existing machine learning-based studies on child malnutrition in India use individual- or household-level microdata and relatively few focus on district-level prediction. District-level modelling is important because policy actions and programme implementation in India are often planned and monitored at the district level. Furthermore, while ensemble learning methods such as Random Forest and XGBoost are increasingly used, the application of meta-heuristic optimization techniques, such as Particle Swarm Optimization (PSO), in the context of child malnutrition prediction remains limited. Studies such as Pandey, S., Sharma, J., & Ali, M. (2024) have applied stepwise logistic regression and regression models.

The present study builds upon this literature by using district-level NFHS-5 data to predict child stunting and wasting through multiple machine learning models and further improves model performance using PSO-based hyperparameter optimization. By doing so, it contributes to both the public health and machine learning literature and provides a data-driven framework for district-level nutritional assessment in India.

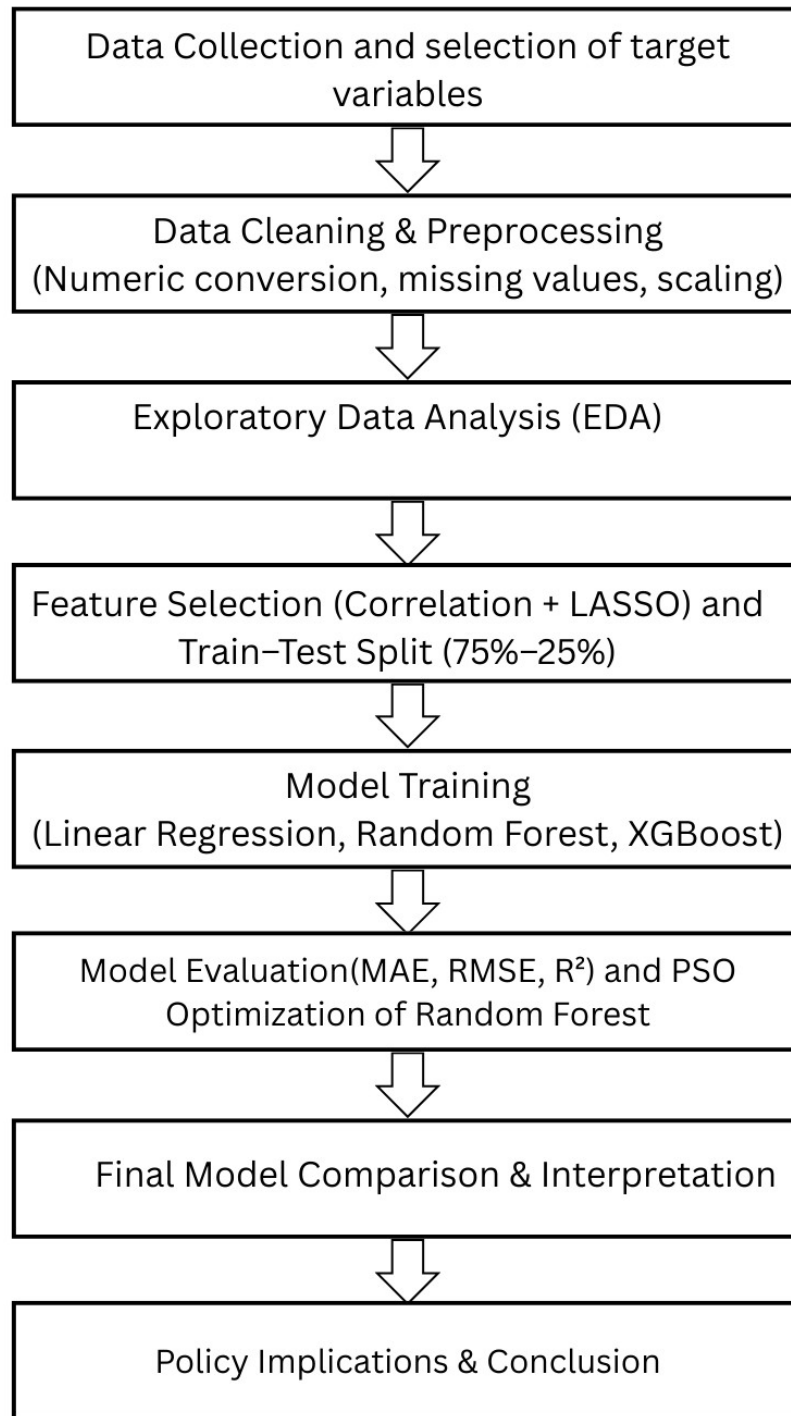


Figure 1 : Flowchart of proposed work

Data Description

This study uses data from the National Family Health Survey (NFHS-5), 2019–20, India. NFHS is a nationally representative survey conducted under the supervision of the Ministry of Health and Family Welfare, Government of India. It provides reliable information on

population health, nutrition, maternal and child health and socio-economic conditions. For this study, the District Fact Sheet dataset of NFHS-5 was used. The dataset contains district-level aggregated indicators for all major health and demographic outcomes across India. After data cleaning, the final dataset consists of 706 districts with 109 indicators for each district.

The study investigates two critical forms of malnutrition in children under five years old: stunting (height-for-age), which indicates chronic or long-term undernutrition and wasting (weight-for-height), which signals acute or short-term undernutrition. To analyse these distinct issues effectively, separate machine learning models were developed for each outcome. From an initial pool of 109 indicators, the models utilize a wide variety of predictors grouped into four main categories to explain malnutrition. These include Household and Environmental Factors (such as sanitation, water and fuel), Maternal Health and Education (including literacy, BMI and antenatal care), Child Health (covering diet, immunization and disease) and broader Socio-Economic indicators (like family planning and health insurance). Together, these variables capture the complex web of biological, environmental and social factors affecting child health.

The analysis uses cross-sectional data from the NFHS-5 dataset, aggregated at the district level. Since the data points represent district averages or percentages rather than individual people, the study uses regression modeling to explain variations between districts. Therefore, the findings should be interpreted as district-level associations rather than direct causal predictions for individual households or children.

Data Preprocessing

The raw NFHS-5 district dataset contained several non-numeric and descriptive variables that were not suitable for machine learning analysis. As a first step, the following preprocessing steps were carried out:

The district name and state/UT name columns were removed, as they do not serve as predictors in regression modelling. All variables expressed as percentages were converted into numeric format. Missing values were handled using mean imputation, as the proportion of missing values was small and the data were continuous in nature. The dataset was checked for duplicate records and none were found. Basic outlier inspection was performed using boxplots and interquartile range (IQR) checks. Since most variables represent district-level percentages, extreme values were retained as they reflect real regional disparities rather than data errors. After preprocessing, the final working dataset consisted of 706 districts and 109 cleaned numeric variables.

Basic descriptive statistics were computed to understand the distribution of key variables, especially the two outcome variables child stunting (%) and child wasting (%).

The district-wise prevalence of stunting showed a wide range, indicating substantial regional variation in chronic undernutrition. Similarly, wasting also varied considerably across districts, reflecting differences in short-term nutritional stress and disease environment.

Measures such as mean, standard deviation, skewness and kurtosis were examined for the main nutrition indicators. The distributions of both stunting and wasting showed slight

skewness, suggesting that while many districts cluster around moderate values, a smaller number of districts experience very high levels of malnutrition.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to examine relationships between child malnutrition and key explanatory variables.

(a) Univariate Analysis

Histograms and boxplots were used to study the distribution of important indicators such as: Stunting and wasting prevalence, Maternal BMI, Household sanitation coverage, Access to improved drinking water

These plots revealed large inter-district variation in both nutritional outcomes and living conditions.

(b) Bivariate Analysis

Scatter plots and correlation analysis were used to examine pairwise relationships between selected predictors and the outcome variables. The analysis showed that: Districts with higher sanitation coverage and female literacy generally had lower stunting levels. Maternal undernutrition and child anaemia were positively associated with higher wasting prevalence. Indicators related to exclusive breastfeeding and adequate diet showed negative associations with wasting.

A correlation heatmap was also used to identify variables that were strongly related to stunting and wasting, which helped in preliminary feature screening.

Since the dataset contained variables measured at different numeric scales, feature scaling was applied before training the machine learning models. Standardization using `StandardScaler` was used so that all predictors had comparable ranges. This step was essential for algorithms such as LASSO regression and XGBoost, which are sensitive to the scale of input variables.

Feature Selection

The NFHS-5 district dataset contains a large number of indicators, many of which are highly correlated with each other. Using all variables directly in machine learning models can lead to problems such as multicollinearity, overfitting and reduced interpretability. Therefore, a systematic feature selection process was carried out to identify the most informative predictors for child stunting and wasting.

Correlation-Based Screening

As a first step, Pearson correlation analysis was performed between all predictor variables and the two target variables stunting and wasting. This helped in identifying variables that had a strong linear association with the outcomes.

The analysis showed that variables related to: Child undernutrition indicators are Maternal BMI, Women's education, Household sanitation and Anaemia prevalence were among the

most strongly correlated with both stunting and wasting. This step provided an initial understanding of which variables were likely to be important for prediction and interpretation.

LASSO Regression for Feature Selection

To further refine the feature set and avoid redundancy, LASSO (Least Absolute Shrinkage and Selection Operator) regression was applied. LASSO is a regularization technique that shrinks the coefficients of less important variables towards zero and effectively removes irrelevant predictors from the model.

Separate LASSO models were fitted for: Stunting and Wasting using standardized predictor variables. The strength of regularization was selected using cross-validation. As a result, only those variables with non-zero coefficients were retained in the final feature set.

This approach allowed the study to: Reduce dimensionality, Control model complexity and Improve generalization performance.

Final Selected Feature Set

After correlation screening and LASSO-based selection, a final set of approximately 28 key predictors was retained for model training. These variables mainly belonged to the following domains:

- Maternal nutrition and health (maternal BMI, maternal anaemia)
- Women's education and empowerment
- Household sanitation and living conditions
- Child feeding practices and anaemia
- Health service utilization and demographic factors

The selected features were found to be consistent with existing public health literature on the determinants of child malnutrition. These variables were then used as inputs for all subsequent machine learning models.

Methodology

This study follows a structured machine learning framework to predict district-level child stunting and wasting using NFHS-5 data. The methodology consists of model formulation, data splitting, model training using multiple algorithms, performance evaluation and hyperparameter optimization using Particle Swarm Optimization (PSO).

Two separate regression-based prediction models were developed:

- **Model 1:** Prediction of district-level child stunting (%)
- **Model 2:** Prediction of district-level child wasting (%)

Both models were trained using the same set of selected predictors obtained from the feature selection stage. Since the outcome variables are continuous percentages, the problem was treated as a regression task.

Machine Learning Algorithms Used

To compare predictive performance across different levels of model complexity, three regression algorithms were applied:

(a) Linear Regression

Linear Regression was used as a baseline model due to its simplicity and high interpretability. It assumes a linear relationship between the predictors and the outcome variable and provides a benchmark to assess the performance of more complex models.

(b) Random Forest Regression

Random Forest is an ensemble learning method that combines predictions from multiple decision trees. It captures non-linear relationships and interactions among variables and is less sensitive to noise than single decision trees. Due to its strong predictive ability, Random Forest was also selected for further optimization using PSO.

(c) XGBoost Regression

XGBoost is a gradient boosting algorithm that builds trees sequentially and improves prediction by minimizing errors at each step. It is known for high predictive accuracy and efficiency, especially in structured data problems.

To evaluate the generalization performance of the models, the dataset was randomly divided into:

- 75% training data (529 districts)
- 25% testing data (177 districts)

The same split was used for both stunting and wasting models to ensure fair comparison across algorithms. All models were trained using only the training set and final performance was evaluated on the test set.

Evaluation Metrics

Model performance was assessed using standard regression evaluation measures:

- **Mean Absolute Error (MAE):** Measures the average absolute prediction error.
- **Root Mean Squared Error (RMSE):** Penalizes larger errors and reflects overall prediction accuracy.
- **Coefficient of Determination (R^2):** Indicates the proportion of variance explained by the model.

These metrics were computed separately for both training and testing data in order to assess model accuracy as well as overfitting.

Particle Swarm Optimization (PSO)

To further improve the performance of the Random Forest model, Particle Swarm Optimization (PSO) was applied for hyperparameter tuning. PSO is a population-based meta-heuristic optimization technique inspired by the collective movement of birds and fish.

In the PSO framework:

- Each particle represents a candidate set of Random Forest hyperparameters.
- The fitness function was defined as the cross-validated RMSE on the training data.
- Particles iteratively updated their positions based on: Their own best performance (personal best) and the best performance among all particles (global best).

The following Random Forest hyperparameters were optimized using PSO:

- Number of trees (n_estimators)
- Maximum tree depth (max_depth)
- Minimum samples required for splitting (min_samples_split)
- Maximum number of features used for splitting (max_features)

After optimization, the best-performing hyperparameter set was used to retrain the Random Forest model and its performance was evaluated on the test data for both stunting and wasting.

Results

This section presents the predictive performance of the machine learning models developed for district-level child stunting and wasting. The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 on both training and testing datasets.

Baseline Linear Regression Results

Linear Regression was used as the baseline model for both outcomes.

For **child stunting**, the Linear Regression model achieved a **test R^2 of 0.61** with a **test RMSE of 5.17**. This indicates that about 61% of the variation in district-level stunting was explained by the selected predictors. The training and testing errors were relatively close, showing that the model did not suffer from serious overfitting.

For **child wasting**, the Linear Regression model performed better, with a **test R^2 of 0.78** and a **test RMSE of 2.78**. This suggests that wasting is more predictable at the district level than stunting. The model showed strong generalization ability with little overfitting.

These results indicate that even a simple linear framework can capture a large share of the variation in child malnutrition at the district level.

Random Forest Results (Before Optimization)

The Random Forest model was then applied to capture non-linear relationships among predictors.

For **stunting**, Random Forest achieved a very high **training R^2 of 0.95**, but the **test R^2 was only 0.62780**, with a **test RMSE of 5.08**. This shows that while the model fitted the training data extremely well, the improvement over Linear Regression on the test data was small, indicating the presence of overfitting.

For **wasting**, Random Forest achieved a **training R^2 of 0.98** and a **test R^2 of 0.76**, with a **test RMSE of 2.91**. Although the training performance was excellent, the test performance was slightly weaker than that of Linear Regression. This again suggests overfitting in the Random Forest model.

XGBoost Results

XGBoost, a powerful boosting-based algorithm, was also applied.

For **stunting**, XGBoost produced a **training R^2 of 0.99**, but the **test R^2 remained at 0.63**, with a **test RMSE of 5.08**. This confirms that XGBoost almost perfectly memorized the training data but did not lead to meaningful improvement in generalization beyond Random Forest.

For **wasting**, XGBoost achieved a **test R^2 of 0.78** with a **test RMSE of 2.81**, which was very close to the Linear Regression performance but again showed strong overfitting in the training phase.

Overall, both Random Forest and XGBoost showed very high training accuracy but limited gains on test data, especially for stunting.

PSO-Optimized Random Forest Results

To improve model generalization, Particle Swarm Optimization (PSO) was applied to tune the hyperparameters of the Random Forest model.

For **stunting**, PSO improved the **test RMSE marginally from 5.08 to 5.07** and the **test R^2 increased slightly from 0.62780 to 0.63**. The improvement was small and some degree of overfitting still remained.

For **wasting**, PSO reduced the **test RMSE from 2.91 to 2.88** and increased the **test R^2 from 0.76 to 0.77**. Although the improvement was modest, PSO slightly reduced overfitting and improved generalization for wasting.

Comparison of Models

Overall, the results show that:

- **For stunting**, the best-performing model in terms of test accuracy was the PSO-optimized Random Forest ($R^2 \approx 0.63$), but Linear Regression remained the most stable and interpretable model.

- For wasting, Linear Regression achieved the highest test R^2 (≈ 0.78), followed closely by XGBoost and PSO-optimized Random Forest.
- Ensemble models such as Random Forest and XGBoost showed strong overfitting, especially for stunting.
- PSO provided small but consistent improvements, particularly for the wasting model.

These findings indicate that wasting is easier to predict than stunting at the district level, likely because wasting is influenced by short-term health and nutritional conditions, while stunting reflects long-term structural deprivation.

Table 1: Model performance for stunting

Model	Train MAE	Test MAE	Train RMSE	Test RMSE	Train R^2	Test R^2
Linear Regression	3.3441	4.0896	4.2702	5.1734	0.7459	0.6143
Random Forest	1.4286	4.0614	1.8255	5.0823	0.9536	0.6278
XGBoost	0.67	4.0875	0.8778	5.0815	0.9893	0.6279
RF + PSO	1.4254	4.0705	1.8192	5.0705	0.9539	0.6295
SVR	3.8793	4.5392	5.1528	5.6317	0.6299	0.5429
SVR+PSO	2.10534	3.7837	3.2744	4.8436	0.8505	0.6619

Table 2: Model performance for wasting

Model	Train MAE	Test MAE	Train RMSE	Test RMSE	Train R^2	Test R^2
Linear Regression	1.8444	2.2056	2.3264	2.7838	0.8781	0.7813
Random Forest	0.7684	2.2859	0.9936	2.9092	0.9778	0.7611
XGBoost	0.3871	2.2217	0.5	2.8069	0.9944	0.7776
RF + PSO	0.8075	2.2532	1.0893	2.8811	0.9733	0.7657
SVR	2.2108	2.4706	3.1510	3.1194	0.7763	0.7253
SVR+PSO	1.8478	2.2010	2.3828	2.7826	0.8720	0.7814

Discussion and Interpretation

This study applied multiple machine learning models to predict district-level child stunting and wasting using NFHS-5 data. The results show clear differences in the predictability of the two forms of malnutrition and provide important insights into their underlying drivers.

Why Wasting Is More Predictable Than Stunting

One of the main findings of this study is that wasting is predicted with higher accuracy than stunting across all models. The best test R^2 for wasting was close to 0.78, whereas the best test R^2 for stunting was about 0.63.

This difference is expected from a public health perspective. Wasting reflects short-term nutritional stress and recent illness, such as diarrhoea, fever and poor feeding practices. These short-term factors are well captured in survey data and therefore respond better to predictive models.

In contrast, stunting reflects long-term and cumulative nutritional deprivation, which is influenced by persistent poverty, intergenerational malnutrition, poor early-life nutrition and long-term environmental conditions. Such deep structural factors are more difficult to capture accurately using cross-sectional district-level data, which explains the relatively lower predictability of stunting.

Overfitting in Non-Linear Models

Both Random Forest and XGBoost achieved very high training accuracy for stunting and wasting but showed limited improvement in test performance compared to Linear Regression. This indicates the presence of overfitting, where the models learn the noise and specific patterns of the training data but fail to generalize well to unseen districts.

This result highlights an important point:

More complex machine learning models do not always guarantee better real-world prediction, especially when the sample size is moderate and the underlying relationships are driven by long-term social and demographic processes.

Role of PSO in Improving Model Performance

Particle Swarm Optimization (PSO) was applied to tune the hyperparameters of the Random Forest model. For both stunting and wasting, PSO led to small but consistent improvements in test RMSE and R^2 . The improvement was slightly stronger for wasting than for stunting.

This suggests that meta-heuristic optimization can help improve generalization by controlling model complexity, but its impact remains limited when the outcome itself is driven by long-term structural factors, as in the case of stunting.

Interpretation of Key Predictors

The feature selection and model interpretation results consistently highlighted several important predictors:

- **Maternal nutrition (BMI and anaemia):** Districts with higher maternal undernutrition showed higher levels of both stunting and wasting, confirming the strong biological link between maternal and child health.
- **Household sanitation:** Poor sanitation emerged as a major predictor of stunting, supporting the well-established link between repeated infections and chronic growth failure.

- **Women's education:** Higher female literacy and schooling were associated with lower stunting and wasting, reflecting better child care practices and improved use of health services.
- **Child anaemia and illness:** These factors were more strongly linked with wasting, highlighting the role of short-term health shocks in acute malnutrition.

These results are consistent with existing public health and nutrition literature and strengthen the validity of the findings.

Economic and Policy Interpretation

From an economic perspective, stunting represents a form of long-term human capital loss, as it affects cognitive development, schooling outcomes and future productivity. The difficulty in predicting stunting using short-term indicators reflects the deep-rooted nature of poverty and deprivation in many districts.

Wasting, on the other hand, reflects short-term vulnerability and is more responsive to immediate health and nutrition interventions. The higher predictability of wasting suggests that timely district-level actions related to child feeding, disease control and maternal care can lead to quicker improvements.

Policy Implications

The findings of this study point to some clear priorities for child nutrition policy and district-level planning in India. Since the analysis is based on district-level data, the results are especially useful for designing targeted, location-specific interventions. Maternal health emerges as a crucial factor, with maternal BMI and anaemia showing a strong link to both stunting and wasting among children. This underlines the importance of strengthening maternal nutrition through iron and folic acid supplementation, better nutrition support for adolescent girls and improved health care during pregnancy. Sanitation and access to clean drinking water also play a major role, as poor living conditions increase children's exposure to infections that interfere with nutrient absorption and long-term growth. Districts with low sanitation coverage should therefore remain a key focus under WASH-related interventions.

Women's education and empowerment also stand out as powerful long-term determinants of child nutrition. Mothers with better education are more likely to follow healthy feeding practices, seek timely medical care and ensure full immunization for their children. At the same time, child health factors such as anaemia, illness, breastfeeding and complementary feeding are strongly linked with wasting, showing the need for timely treatment of infections, promotion of exclusive breastfeeding, improved feeding after six months and regular screening for child anaemia. Finally, this study demonstrates that machine learning models can be effectively used for district-level monitoring of child malnutrition. Even simpler models performed well, suggesting that data-driven tools can help policymakers identify high-risk districts and allocate resources more efficiently.

Limitations

While this study offers useful insights into district-level prediction of child stunting and wasting in India, it has a few important limitations. Since the analysis is based on aggregated district-level NFHS-5 data, the models reflect average patterns and cannot explain individual or household-level outcomes or establish causal relationships. The cross-sectional nature of the data also prevents the analysis of time trends and seasonal variations, which are especially relevant for wasting. In addition, some overfitting was observed in more complex models like Random Forest and XGBoost, suggesting that predictions may be sensitive to the data structure. Despite these constraints, the study still provides a strong district-level predictive framework using nationally representative data and remains valuable for policy-oriented analysis.

Conclusion

This study developed a district-level machine learning framework using NFHS-5 (2019–21) data to predict child stunting and wasting in India through a structured pipeline of preprocessing, feature selection, model comparison and optimization. The results show that wasting is more accurately predictable than stunting at the district level. Linear Regression performed best for wasting with a test R^2 of about 0.78, while stunting was best predicted by the PSO-optimized Random Forest with a test R^2 of about 0.63. This difference reflects the short-term nature of wasting compared to the long-term structural roots of stunting.

Although complex models like Random Forest and XGBoost showed very high training accuracy, their performance on test data was only marginally better than simpler models and in some cases weaker for wasting. Particle Swarm Optimization led to small but consistent improvements and helped reduce overfitting slightly. Key predictors included maternal nutrition, sanitation, women's education, child anaemia and feeding practices, highlighting the need for integrated interventions. Overall, the study shows that machine learning can be a useful tool for district-level monitoring of child malnutrition, helping identify high-risk areas and guide targeted policy actions.

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